

PollGrid

A real-time US political intelligence platform: primary-source polling, campaign finance, official election results, legislative activity, and campaign rhetoric — mapped against hyper-granular demographics from the nation down to the precinct, correlated across races and back to 1950, reasoned over by a local Ollama-powered Analyst that never touches the actual math, and rendered on a hand-built 3D map.

Everything below assumes a from-scratch build. Nothing in the current repo is treated as load-bearing.

50+6 STATES, DC & TERRITORIES	5 GEOGRAPHIC TIERS, NATION → PRECINCT	3,143+ COUNTIES, PARISHES & BOROUGHs	435 HOUSE DISTRICTS, REDISTRICTING-VERSIONED	2 STATES SPLITTING ELECTORS BY DISTRICT (ME, NE)	1789- PRESIDENTIAL RETURNS IMPORTED, CONFIDENCE-TAGGED BY ERA
0 LLM CALLS IN THE AVERAGING/FORECAST MATH PATH	0 AUTO-PUBLISHED RACE CALLS, EVER				

Vision & hard constraints | Architecture at a glance | Domain model | Ingestion layer | Ingestion, made concrete | Extraction & the fact chain | The intelligence engine

Vision & hard constraints

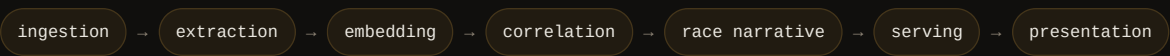
The pitch: take the actual engineering that makes GlobeGrid good — thread-per-source ingestion that degrades instead of crashing, a permanent hash-chained fact chain, correlation across sources *and* across decades, a bounded LLM that narrates but never calculates, a hand-rolled WebGL2 renderer with real LOD, a buildless frontend with real sound design and real theming — and point all of it at US elections instead of world events. Not a redesign of the old PollGrid spec. A new one, built the way GlobeGrid was actually built.

§ Non-negotiables

1. **Primary sources only.** Individual pollster releases, the FEC API, Census, the AP Elections API, state Secretary of State feeds, Congress.gov, official campaign transcripts and press releases. Never scrape RealClearPolitics / 538 / 270ToWin's aggregated numbers, ratings, or prose — link out to them, build every average from the underlying releases ourselves.
2. **Race calls are never automated.** The election-night engine flags a race `CALLABLE` for human review and stops there. `election_night.auto_publish_calls` is hardcoded `False` in code, not just config, and `race_calls.called_by` must be a real human identifier — never `"system"`, never a model name.
3. **Append-only, everywhere.** Poll averages, fundamentals snapshots, forecasts, and the volatility index are all dated snapshots on an `as_of` axis, never overwritten in place — the same mechanism that runs the archive back to 1950 also runs the time-capsule scrubber for a race that happened yesterday.
4. **Provenance is hash-chained, not just logged.** Every extracted fact and every prediction chains to the previous row's hash. It proves our own copy wasn't altered after being written — it does not, and never claims to, prove the underlying poll or statement was true.
5. **The LLM is bounded and provider-agnostic.** Local-first (Ollama), cloud fallbacks behind it. It writes race narratives, dossier prose, and rhetoric classifications inside a fixed rubric — it never touches poll averaging, the fundamentals model, forecasting math, correlation thresholds, the volatility index, or a race call. Every generative feature ships a deterministic fallback the pipeline actually exercises on failure.
6. **Every tunable lives in one config file.** Thresholds, intervals, weights, model names — `config.yaml` + `.env`, schema-validated at boot, fails the whole process loudly on a bad value rather than limping forward.
7. **A failing source degrades, never crashes.** Thread-per-source, exponential backoff capped at a config ceiling, three-state health (`ok` / `degraded` / `down`) — one dead pollster feed can't take down election night.
8. **Synthetic data is always tagged and always purgeable in one operation.** `is_synthetic=1`, one command removes every trace.

Architecture at a glance

The same seven-stage shape, aimed at a different domain.



Backend: stdlib-first Python — `http.server` + a hand-rolled router, a hand-rolled WebSocket, SQLite in WAL mode, daemon-thread scheduler. No async framework, no ORM, no Postgres. Frontend: buildless ES modules, a hand-rolled WebGL2 renderer, no React/Vite/npm/CDN. The zero-install pledge carries over in full — `python run.py` and it's up, database created, sources seeded, ingestion running.

LAYER	WHAT IT OWNS
core/	Config loader, SQLite session/locking, hash-chain provenance, gazetteer
ingestion/	Scheduler + one adapter per source type (polls, FEC, Census, AP, state SoS, Congress, transcripts, news, markets, social)
modeling/	Averaging, fundamentals, ideology, forecasting, chamber simulation, correlation, volatility, second-order links, coalition detector, counterfactual, corroboration, rhetoric/NLP, race-calling, pollster ratings, redistricting fairness, the audit trail
domain/	Candidates (every historical filer), parties, races, geography (states → districts → county-equivalents → precincts) and demographics at every tier, the curated-floor + sync + AI-fill knowledge layer
analyst/	Context-pack assembly + the Ollama-first reasoning/Q&A engine
api/	Router, REST routes, WebSocket feed, static frontend serving — every read route accepts <code>as_of</code>
frontend/	Buildless SPA — tiered map, the client-side map builder, a global polls window, race/geography pages with electoral history, the Analyst, counterfactual UI, time capsule
scripts/	Offline boundary/redistricting build pipeline, seeds, historical backfill importers

Domain model

The geopolitics-knowledge-layer trick, aimed at candidates instead of countries — and pushed one tier deeper than GlobeGrid ever needed to go.

§ Geography, five tiers deep

Nation → State → Congressional District → county-equivalent → precinct. Every tier is a real, separately-modeled entity, not a label pasted onto the one above it — because the naming and the electoral math both genuinely change tier to tier.

TIER	SCOPE	WHAT'S TIER-SPECIFIC ABOUT IT
Nation	1	National polling averages, national fundamentals, the generic ballot.
State	50 + DC + 5 territories	DC and the territories carry no voting House seat and are tracked for context, not electoral math. Every state row carries an <code>elector_method</code> field.
Congressional district	435 voting + non-voting delegates	Redistricting-versioned (<code>effective_from</code> / <code>effective_to</code>) so the 2012-cycle lines and the 2022-cycle lines are genuinely different geometries, never the same shape relabeled.
County-equivalent	3,143+	Modeled with a <code>type</code> field — <code>county</code> / <code>parish</code> (Louisiana) / <code>borough</code> & <code>census_area</code> (Alaska) / <code>independent_city</code> (Virginia) — never hardcoded as "county" everywhere else in the schema.
Precinct / VTD	~170,000, coverage varies by state	Boundaries and results from VEST-style precinct projects plus county election-office GIS releases where published. Genuinely missing in some states — flagged <code>thin_coverage</code> , never silently backfilled.

ELECTORAL VOTE ALLOCATION — MODELED EXPLICITLY, NEVER ASSUMED

48 states plus DC award electors winner-take-all at the state level. **Maine and Nebraska don't** — one elector per congressional district's winner, plus two statewide electors for the overall state winner. This isn't a special case bolted on later: every state carries an `elector_method` of `winner_take_all` or `congressional_district` , and the presidential model reads its electoral math off the district tier for exactly those two states and off the state tier for the other 48 + DC.

§ Demographics, at every tier

The same discipline GlobeGrid applied to country stats — "sourced, never LLM-guessed" — applied here at five tiers instead of one. A single polymorphic `demographics` table (keyed by tier + entity + `as_of`) carries every category below for the nation, every state, every congressional district, every county-equivalent, and every precinct coverage allows:

CATEGORY	INCLUDES	SOURCE
Population & age	Total population, voting-age population, citizen voting-age population, median age, age brackets	Census ACS / decennial
Race & ethnicity	Full Census race/ethnicity breakdown, and its trend over time	Census ACS / decennial
Education	Attainment brackets — less than HS, HS, some college, bachelor's+, graduate	Census ACS
Economic	Median household income, income distribution, unemployment rate, industry mix	Census ACS / BLS
Housing & urbanicity	Homeownership rate, housing cost burden, urban/suburban/rural classification	Census ACS / USDA Rural-Urban Continuum Codes
Social & nativity	Foreign-born share, language spoken at home, veteran status, disability rate	Census ACS
Political-historical	Party registration mix where states register by party, turnout rate, past results back to 1950	State voter files / the certified-results archive

Every row carries a `confidence` tag, not just a value. State, district, and county-equivalent figures are `measured` — direct Census/BLS estimates. Precinct-level figures are `derived` — apportioned from Census block data by areal interpolation, since no federal agency publishes precinct-level demographics directly — and the UI is required to show that

distinction every time it renders a precinct number, the same honesty a "thin coverage" badge enforces elsewhere.

§ Races covered, for now

President, Senate (Class I/II/III tracked), Governor, and House (435 + non-voting territorial delegates) — plus primaries as their own race-phase with their own candidate set. That's the whole scope of this build. **Ballot measures and state legislative chambers are deliberately not in it** — the schema leaves the room (a race can already be typed generically), but nothing ingests, models, or surfaces either until state houses are explicitly back in scope.

§ Entities

Candidates & officeholders

Bio, party, current office, campaign finance summary, key positions, a deterministic ideology score — never an LLM guess at where someone sits on a spectrum.

Parties

National committees + all 50 state parties, platform positions.

Money & orgs

PACs, Super PACs, 527s, aggregated disclosed donors, endorsing interest groups.

Media

Outlets carry a reliability/leaning tier, same idea as GlobeGrid's source tiers — feeds the media-framing matrix and keeps low-signal outlets from drowning out wire-quality sources.

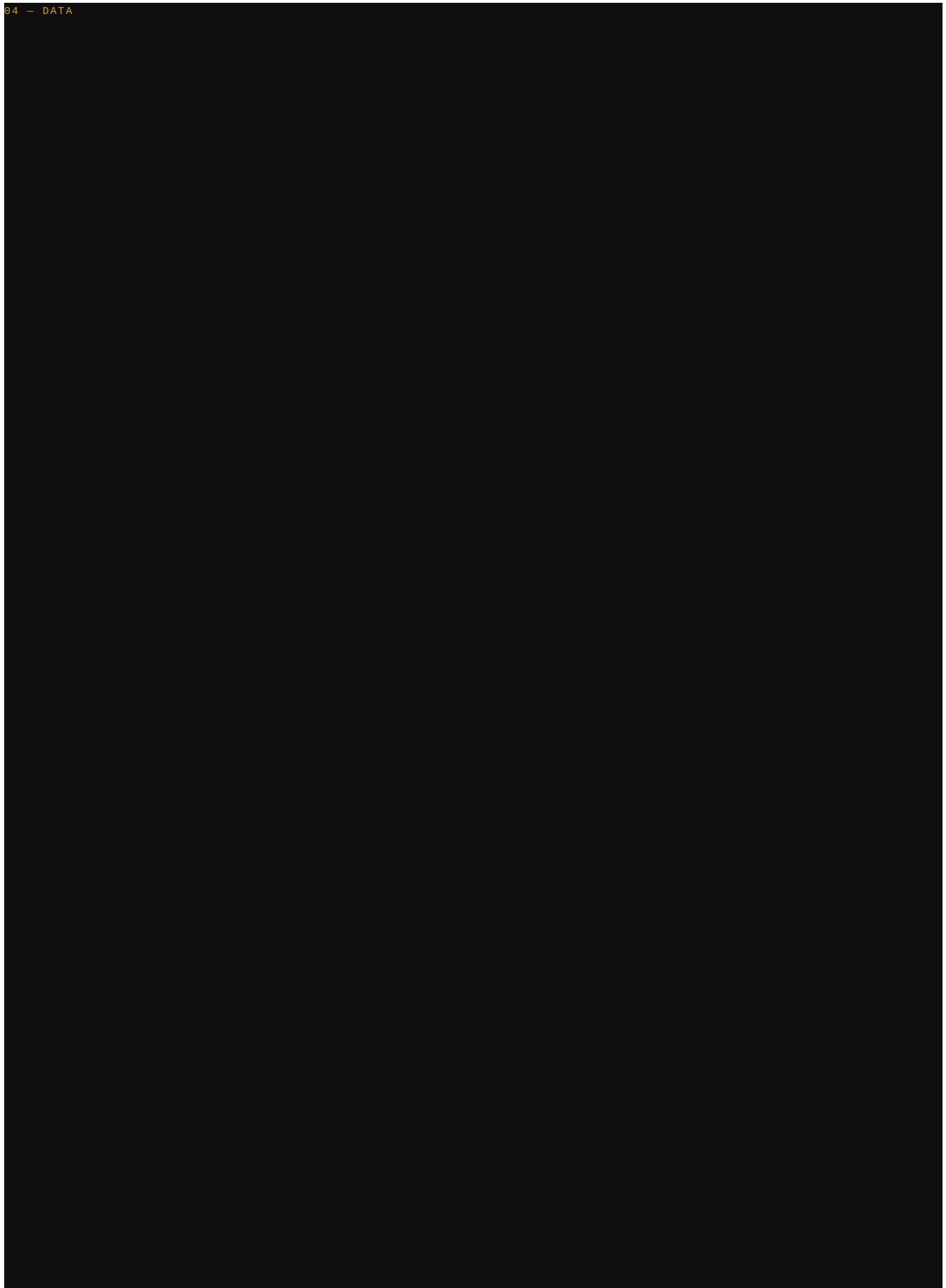
§ The three-tier acquisition pattern

Directly lifted from GlobeGrid's country/leader/party layer, because it's the single best idea in that codebase — now doing double duty for candidates *and* for the demographic panel underneath them:

1. **Curated floor.** Every sitting member of Congress, every current governor, and every major-party presidential/Senate/gubernatorial candidate gets a hand-seeded row with cited sources — so a candidate page is never blank on day one.
2. **Free structured-API sync.** FEC (finance), Census (demographics at every tier), Congress.gov (votes, bill sponsorship) refresh on a schedule, purely additive — **COALESCE** -style, never overwriting a curated value, and once a sync has touched a row the seed step leaves it alone.
3. **LLM fills the remaining gaps** — bio prose, a plain-language policy-position summary, a party-history paragraph — generated in the background, cached, merged over the floor, never blocking a page from opening.

Coverage isn't limited to the majors. Every candidate who has ever appeared on a general-election ballot for President, Senate, Governor, or House gets a real page — a stub the instant an official filing record or historical-archive import creates the row, thickening automatically as sync and AI-fill reach it. A losing 1962 House candidate and a sitting Senator both get a page; they just don't get the same depth on day one, and the page says so.

04 — DATA



Ingestion layer

One daemon thread per source, three-state health, a budget counter for anything rate-capped.

The scheduler spawns one daemon thread per active source and re-reads that source's own database row every tick, so toggling a source off or editing its poll interval takes effect on the next cycle with no restart. Each thread waits on a shared stop-event rather than a plain sleep, so shutdown is immediate. Backoff is exponential, capped at a configured ceiling, same as the ground rules state.

`ingestion/pollsters/*.py`

Polling

One adapter per pollster feed/API — start with the dozen-plus highest-quality, methodologically-transparent shops, expand over time. Normalizes into one `Poll` shape: candidate, race, sample size, margin of error, field dates, population (LV/RV/A), methodology.

`ingestion/fec.py`

Campaign finance

Contributions, independent expenditures, committee filings straight from the FEC API — the deterministic input to the fundraising-trend signal used by both fundamentals and corroboration.

`ingestion/census.py`

Demographics

ACS + decennial data per district/county on its own release cadence — a fundamentals-model input, never guessed.

`ingestion/ap_elections.py`

Official results

The ground-truth "sensor" — the one feed nothing else is allowed to override, and the only thing election night is allowed to call a race against.

`ingestion/state_sos/*.py`

State certification feeds

One adapter per state Secretary of State results/certification feed — start with competitive-state coverage, expand every cycle.

`ingestion/congress_gov.py`

Legislative record

Roll-call votes, bill sponsorship, committee assignments — feeds ideology scoring and the rhetoric/topic layer.

`ingestion/transcripts.py`

Campaign speech

Official press releases, rally remarks, debate transcripts from campaign/official sources only — text input to the rhetoric NLP layer.

`ingestion/rss.py`

Political news

Wire services and political desks, generic RSS/Atom parsing, same shape as any GlobeGrid-style news adapter.

`ingestion/markets.py`

Prediction markets

Legal markets (Kalshi, and others as availability allows) feeding the forecast/market-divergence signal.

`ingestion/social.py`

Social signal

Low-reliability-tier volume/sentiment only — never a source of fact, only of noise-aware context.

§ Failure isolation

OUTCOME	HEALTH	NEXT INTERVAL
Success	ok , failure counter reset	base interval
Missing API key	degraded forever	flat, capped — never escalates
Daily budget exhausted	ok — deliberately not a failure	flat, capped
Any other exception	degraded , then down after N failures	exponential, capped

Ingestion, made concrete

Real endpoints, real variable IDs, and an honest tiered strategy for official results — not hand-waving.

§ Census, concretely

`ingestion/census.py` hits the ACS 5-year endpoint — `https://api.census.gov/data/{year}/acs/acs5?get=NAME,{variables}&for={geography}&in={parent}&key={CENSUS_API_KEY}` — the 5-year product, not the 1-year, because 1-year estimates only publish for geographies with 65,000+ population and this platform needs county- and district-level reliability everywhere. Geography clauses differ by tier exactly as the Census API requires: `for=state:*`, `for=county:*&in=state:XX`, and — confirmed against the real API — `for=congressional%20district:*&in=state:XX` for districts. Tract/block-group pulls (`for=tract:*&in=state:XX%20county:YYY`) are the raw input to the precinct-derivation step below, never queried directly by a user-facing feature.

DEMOGRAPHIC CATEGORY	REAL ACS TABLE(S)
Population & age	<code>B01003</code> (total population), <code>B01001</code> (sex by age)
Voting-age / citizen voting-age population	<code>B01001</code> for voting-age brackets; true CVAP is not a standard ACS Detailed Table — it comes from the Census Bureau's separate annual CVAP Special Tabulation product, built specifically for Voting Rights Act analysis. Use that as its own ingestion path, not a B-table.
Race & ethnicity	<code>B02001</code> (race alone), <code>B03002</code> (Hispanic/Latino origin by race — the standard combined table)
Education	<code>B15003</code> (attainment, population 25+)
Economic	<code>B19013</code> (median household income), <code>B19001</code> (income distribution), <code>B23025</code> (employment status)
Housing & urbanicity	<code>B25003</code> (tenure/homeownership); urban/rural classification from the Census Bureau's urban-area delineation files, not an ACS table
Social & nativity	<code>B05002</code> (place of birth/foreign-born), <code>B16001</code> (language at home), <code>B21001</code> (veteran status), <code>B18101</code> (disability status)

THE REAL VINTAGE-MISMATCH PROBLEM — AND HOW IT'S HANDLED

ACS congressional-district data is published against whichever Congress's boundaries were seated during the survey years — when a state redraws districts mid-decade, an ACS release can reflect the *prior* map for a year or more before the Bureau catches up. The fix isn't to ignore this: every Census pull is tagged with the specific Congress vintage it was published against, matched against the `congressional_districts` table's own `effective_from` / `effective_to` boundary versions — a demographic row is only ever joined to the district-shape version it actually describes, never silently mapped onto "whatever CD lines are current now."

Cadence: ACS 5-year estimates refresh annually (a rolling window); the decennial count (`dec/pl` for the PL 94-171 redistricting file, `dec/dhc` for detailed demographic/housing characteristics) lands once every ten years plus the mid-decade redistricting-file release. Boundary geometry itself comes from TIGER/Line shapefiles (already the offline build pipeline's source in [The map](#)) — TIGERweb, the live map-service version of the same data, isn't needed since nothing here calls a live GIS service at runtime.

Precinct-level derivation, named precisely: population-weighted areal interpolation — each Census block's population (from the decennial block-level count, the finest population geography that exists) is distributed into whichever precinct(s) its geometry overlaps, weighted by the fraction of the block's population estimated to fall in each; a precinct's demographic estimate is then the population-weighted sum of the blocks assigned to it. This is a real, named technique with well-understood error characteristics — it degrades when a precinct boundary splits a Census block unevenly (rare, since precincts are usually drawn

from whole blocks, but not guaranteed) — which is exactly why every precinct-level demographic row carries `confidence=derived`, never `measured`.

§ **FEC, concretely**

`ingestion/fec.py` uses the real, free, public OpenFEC API (`api.open.fec.gov`) — `DEMO_KEY` works for development at a throttled rate; a registered key raises the ceiling to 1,000 calls/hour. Endpoints: `/candidates/` and `/candidate/{candidate_id}/` for the roster that seeds every historical and current candidate stub, `/committees/` for PAC/campaign-committee linkage, `/schedules/schedule_a/` for itemized individual contributions, `/schedules/schedule_a/by_contributor/` for aggregated donor-side rollups, and `/schedules/schedule_b/` for itemized spending/disbursements — the deterministic input to both the fundamentals model's fundraising-ratio term and the corroboration engine's finance signal.

§ **Congress.gov, concretely**

`ingestion/congress.gov.py` uses the real, free (API-key-gated) Congress.gov API for bills, sponsorship, and roll-call votes. **Honest caveat:** its House Roll Call Votes endpoint currently only covers the 118th Congress forward (2023–present) — it is not a full historical roll-call archive. For ideology scoring and roll-call history before 2023, the platform pulls from **VoteView** (Poole & Rosenthal's DW-NOMINATE project, public-domain downloads covering every Congress back to the 1st) — a real, load-bearing academic source rather than a gap papered over, and one that hands the ideology-scoring model an actual peer-reviewed methodology to sit on top of instead of inventing its own from scratch.

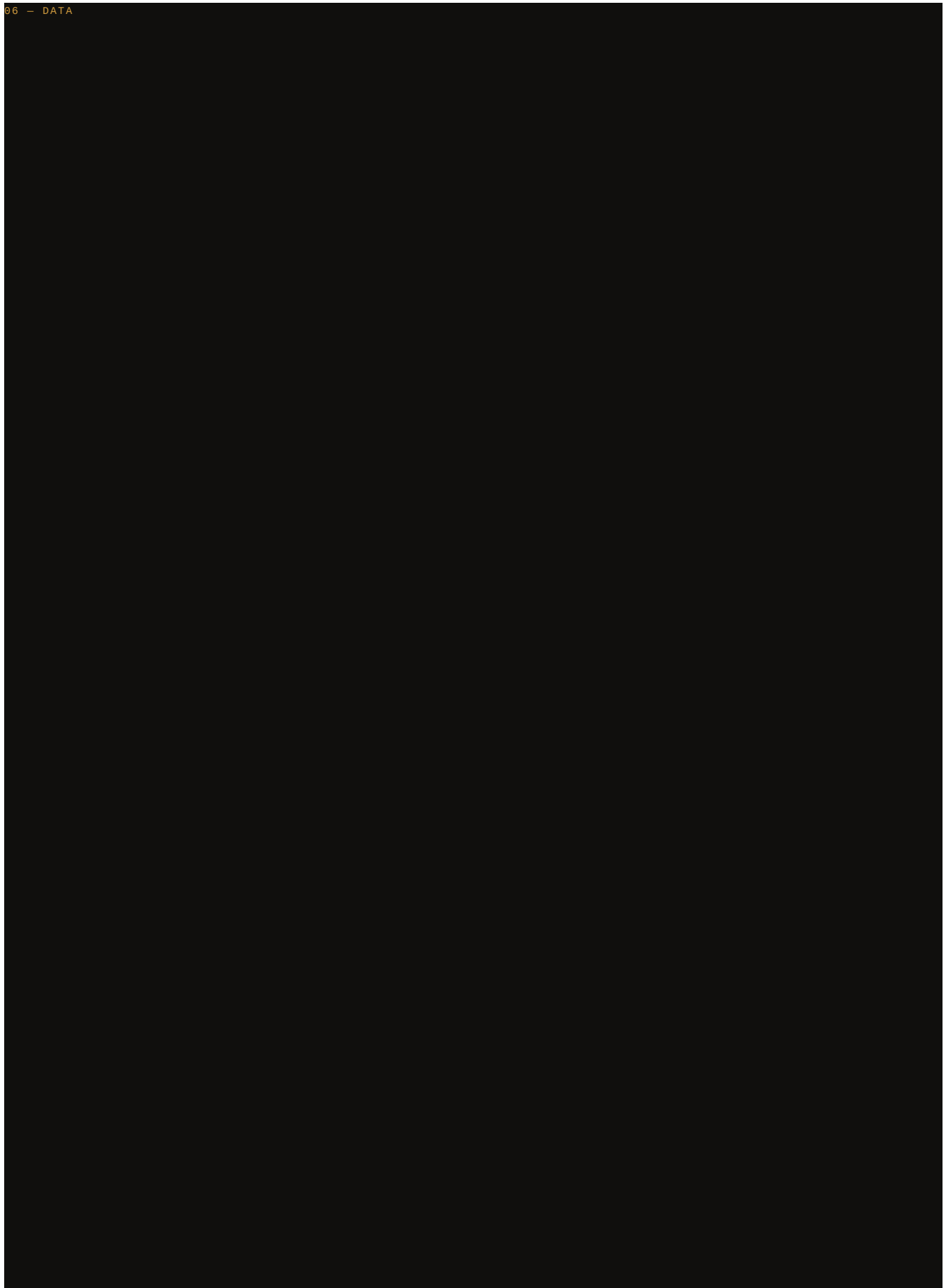
§ **Official results: the honest tiered strategy**

The single biggest access-reality gap in an earlier draft: the AP Elections API (`developer.ap.org/ap-elections-api`) is a **real commercial product** requiring an account and a negotiated license — not a free primary source a small team simply pulls from on day one. Here's the honest, buildable tier structure instead:

TIER	SOURCE	WHAT IT'S ACTUALLY FOR
1 — native, live	Each state's own results feed, for the subset of states that publish genuinely clean machine-readable results (a minority, growing slowly)	Real-time results on election night for those states, ingested the instant they're posted
2 — community, lagged	OpenElections (co-founded 2012 by Serdar Turgoren and Derek Willis; standardized, versioned CSVs covering federal/statewide/state-legislative results) and VEST (University of Florida + Wichita State; the most comprehensive precinct-boundary-plus-results database that exists, distributed via Harvard Dataverse)	The backbone of the deep historical archive and of precinct-level results generally — both projects finalize a cycle weeks to months after election day, which is exactly acceptable for history but not for election night
3 — AP, optional upgrade	The real AP Elections API, gated behind <code>ingestion.ap_elections.enabled</code> (default <code>false</code>)	Turned on only once there's budget to justify it — the UI honestly labels which tier is actually live for a given race ("results via [State] Secretary of State" vs. "results via AP Elections"), never implying AP-grade calling speed from a tier-1 feed that can't deliver it

Module structure mirrors the tiers directly: `ingestion/results/tier1_native/{state}.py` (one file per state with a real feed), `ingestion/results/tier2_openelections.py` (bulk-syncs OpenElections' tagged CSV releases and VEST's per-cycle Harvard Dataverse deposits), `ingestion/results/tier3_ap.py` (the gated, optional adapter), and `ingestion/results/manual_entry.py` — a small authenticated internal tool for the rare jurisdiction with no digital feed at all, every row it produces tagged `source_tier=manual` so its provenance stays visible rather than laundered to look automated.

06 — DATA



Extraction & the fact chain

Polls parse directly. Text gets NLP. Everything gets hash-chained before it's ever stored.

Polls need no NLP at all — they parse straight into a structured `polls` fact. News, speeches, and transcripts go through: entity recognition (spaCy if installed, a deterministic capitalized-run regex fallback otherwise — the same auto-detect-with-fallback idiom as every other optional dependency in this design); a politics-tuned category classifier (`polling` / `finance` / `legislation` / `endorsement` / `scandal` / `debate` / `election-result` / `rhetoric` / `campaign-event` , word-boundary keyword matching with a fixed tie-break order so "war on drugs" doesn't trip a conflict keyword); entity canonicalization (candidate and party name variants resolved via exact-alias cache, then fuzzy match, so "Gov. Whitmer" and "Gretchen Whitmer" and "Whitmer" collapse to one canonical id); gazetteer geocoding to county/district (dateline first, a curated notable-places override for newsworthy-but-small locales, then a Census-Gazetteer-derived place list); and a hash-chain provenance stamp computed *before* insert.

Nothing in `extracted_facts` or `polls` is ever deleted. That table is the permanent record the 1950-archive and the time-capsule scrubber both read from.

07 — INTELLIGENCE

The intelligence engine

Deterministic math for anything that gates a claim. The model only ever narrates.

§ The deterministic core

Poll averaging

Recency-weighted, house-effect-adjusted, sample-size-weighted — every weight config-driven, zero LLM anywhere near it.

Fundamentals model

Incumbency, generic-ballot lean, economic indicators, partisan lean (a computed PVI-style score), fundraising ratio — a weighted composite, not a model's opinion.

Ideology scoring

A deterministic proxy from roll-call votes, donor patterns, and public position statements scored against a fixed rubric.

Volatility index

0–100 composite of poll-movement magnitude, cross-pollster spread, and news volume; z-score plus a CUSUM chart catch a real swing versus noise, same technique as an instability index built for conflict data, retargeted at a polling average.

§ The forecast, and its gate

Forecasting sits on top of the averages and fundamentals, itself deterministic — but a race-type only **earns** visible forecasts after a nightly backtest, replayed against the archive back to 1950, clears a Brier-score ceiling over a minimum number of graded past predictions. That backtest is pure SQL and arithmetic, deliberately not an LLM: no peeking at outcomes that haven't happened yet in the replay, no hindsight. The scorecard shows every race-type's track record, including the ones still failing — the same "the honesty is the moat" principle a serious forecasting product has to earn, not assert.

§ Race narrative generation

One LLM call per race-cluster update: a strict JSON contract — `what_changed` , `why_it_might_have_changed` (grounded in cited facts, never invented), `what_to_watch` , `confidence` . Malformed output gets exactly one retry, then a deterministic templated summary built straight from the numbers. A devil's-advocate pass may only *lower* stated confidence, enforced by a rank comparison in code — never by trusting the model's own claim.

§ Beyond the primary narrative

Second-order links

Two races moving together with no obvious shared cause get flagged for a plausible-common-cause check — a fundraising surge in one district and a poll shift in a neighboring one, say.

Counterfactual engine

"What if this candidate drops out," "what if turnout matches 2020 instead of 2016" — branches grounded via full-text search against the 1950-present archive, each carrying a precedent count from

Demographic coalition detector

Deterministic regression across the demographic panel at whatever tier a race lives in — which variables (education mix, age structure, urbanicity, income) explain the most movement

	real past cycles, honest non-summing probabilities, deepen-on-demand rather than generated to full depth up front.	over the last several cycles. The number comes from statistics; only the plain-English explanation of it comes from the Analyst.
Corroboration A poll's implied direction checked against signals a model can't hallucinate: FEC fundraising trend, campaign-event frequency, ad-spend trend, voter-registration change, early-vote request volume. Agreement across independently-sourced channels raises a corroboration badge on the race page.	Rhetoric & framing A fixed policy-topic taxonomy, stance classification per topic (rubric-constrained LLM with a deterministic keyword fallback), and a media-framing matrix comparing how differently-leaning outlets cover the identical story.	Election night AP and state feeds drive a deterministic CALLABLE flag — margin versus estimated remaining vote, informed by historical county-reporting curves — that a human, and only a human, turns into a call.

The genius layer: quantitative × qualitative

How PollGrid gets smarter than a poll average without ever letting the model touch a number.

Every existing aggregator leaves the same thing on the table. RealClearPolitics-style sites average and stop. 538-style forecasters blend polls with fundamentals — genuinely quantitative, but structurally blind to everything that hasn't shown up in a poll yet: a scandal breaking this morning, a coalition visibly cracking in coverage, a challenger's debate performance landing badly. 270ToWin has no model at all. The obvious fix — "ask an LLM who's going to win" — is exactly what this project's own hard rule forbids, and for good reason: it would produce an unaccountable black box no more trustworthy than a pundit's gut, just faster.

The actual fix is a pipeline, not a shortcut.

§ Step one: the Qualitative Factor Scorecard

Every race carries a fixed, named vector of scores — one per factor in [the factors taxonomy](#) — re-scored by the LLM only when new relevant facts land for that race, cached otherwise. Every score is numeric or ordinal, every score is cited back to the specific ingested fact(s) that justified it, and the scoring runs against a fixed rubric per factor (never open-ended "what do you think happens here"). If no LLM is available, every LLM-scored factor defaults to a neutral 0 — the scorecard degrades to "quantitative-only," never blocks, never guesses. This is a strict generalization of the causal-narrative pattern already established elsewhere in this design: same one-call-per-update discipline, same cited-and-cached rule, same deterministic neutral fallback.

§ Step two: a real, named, auditable combination

The Qualitative Factor Scorecard becomes plain numeric features — nothing more — fed into a **penalized logistic regression (elastic-net) stacking model** alongside the existing deterministic quantitative layer: the poll-average-plus-fundamentals blend's own probability, and the demographic coalition detector's output. This is a real, well-understood statistical-learning technique (a meta-learner over interpretable base predictions, the same family of idea behind any serious ensemble forecast) — chosen specifically because elastic-net regularization shrinks a factor's fitted weight toward zero if it isn't actually earning its keep, rather than a human having to hand-prune a factor that "should" matter but doesn't. **The model itself is refit against the historical archive using the same replay-based backtest machinery that already gates the quantitative-only forecast** — no LLM anywhere near the fitting process, no hindsight, coefficients stored as plain data and fully inspectable.

§ Step three: it has to earn the right to be shown — against its own baseline

THE RULE THAT KEEPS THIS HONEST

A qualitative-augmented forecast for a given race-type category is only ever shown live if it **beats the quantitative-only baseline's Brier score** in the same nightly backtest that already gates every forecast category. The platform is not allowed to prefer the more impressive-sounding model — only the one that's demonstrably more calibrated, category by category. If the qualitative layer doesn't earn its keep for, say, gubernatorial races, gubernatorial forecasts simply run quantitative-only until it does. As more graded Factor-Scorecard-to-outcome pairs accumulate in the historical archive, the elastic-net weights refit periodically and the model can genuinely get more accurate over time — real statistical learning from a growing track record, not a vague claim that "the AI learns."

§ Why this is a structural advantage, not a marketing claim

No existing aggregator does this end to end. 538-style models are quantitative-only by construction — qualitative judgment enters, if at all, as an analyst's ad hoc adjustment, not a systematized, cited, backtested pipeline running across every tracked race. RCP-style sites don't model at all. 270ToWin has no forecast, only a canvas for the user's own assumptions. The novel mechanism here isn't "an LLM helps predict elections" — it's that the LLM's only job is turning an ocean of unstructured text into structured, cited, rubric-scored data at a scale no team of human analysts could sustain across hundreds of races simultaneously, and a transparent, refittable, backtested statistical model does the actual predicting on top of it.

09 — INTELLIGENCE

The factors taxonomy

Every qualitative factor the Scorecard tracks, its real grounding, and exactly how it's measured.

Every row below states plainly whether it's computed **deterministically** from data already in the schema, or **LLM-rubric-scored** from text against a fixed scale — nothing is left ambiguous about which.

GIVEN EXTRA DEPTH, ON REQUEST: GENERATIONAL COHORTS

Grounded in Mannheim's political-generations theory — the political events of one's formative years leave a durable imprint on lifelong partisan lean — and in **generational replacement**, one of the best-documented drivers of long-run partisan change in the discipline: as older cohorts are replaced by younger ones at the ballot box, the electorate's aggregate lean shifts even if no individual voter changes their mind. The platform tracks named cohorts (Silent, Boomer, Gen X, Millennial, Gen Z) by their real, distinct turnout propensities, issue priorities, and media-consumption patterns, computed straight from the age-bracket columns already in the `demographics` table — and maintains a **generational-replacement index** per geography: the trend in post-Boomer share of the electorate over successive cycles, a deterministic leading indicator of long-run drift, tracked separately from any single-cycle factor below.

FACTOR FAMILY	REAL GROUNDING	MEASUREMENT
Retrospective / economic voting	Fiorina's retrospective-voting theory	Deterministic (real income/unemployment/inflation trend) + LLM-rubric (perceived-economy sentiment, which can diverge from the raw indicator)
Anti-incumbency / time-for-change	Abramowitz-style structural midterm models; the president's-party midterm penalty, the "six-year itch"	Deterministic formula on incumbency + midterm-cycle position + approval trend
Generational cohorts & replacement	Mannheim's political generations; generational replacement theory	Deterministic composition + trend (see callout above)
Realignment / secular partisan change	V.O. Key's critical-elections theory; Sundquist's realignment framework	Deterministic — long-run partisan-lean trend line per geography
Valence & candidate quality	Stokes' valence-politics concept	LLM-rubric — competence/authenticity narrative scored from coverage and debate transcripts
Coattail / reverse-coattail effects	Down-ballot correlation with top-of-ticket performance	Deterministic — statistical correlation across same-cycle race results
Intra-party ideological factions	Factional-politics research on primary vs. general electorates	Deterministic clustering (ideology scores, primary vote share) + LLM-rubric (which faction holds the "center of gravity" in current coverage)
Political geography	Elazar's political-culture regions; urban-rural polarization; Bishop's "Big Sort"	Deterministic — urbanicity classification + migration/population-change trend
Religious/denominational blocs	Denominational voting-bloc research (explicitly not treated as monolithic)	Deterministic composition from Pew Religious Landscape estimates (a distinct, honestly-cited source — Census doesn't ask religion) + LLM-rubric for narrative salience only
Ethnic/community voting patterns	Voting-bloc research, with the same explicit non-monolithic caution (e.g. Latino voters vary by national origin, generation, religiosity)	Deterministic composition (race/ethnicity) + LLM-rubric for coverage salience only, never for assuming a bloc's vote
Socioeconomic / "diploma divide"	Class-based educational realignment research	Deterministic — education/income composition trend
Candidate charisma / personality	Valence-politics candidate-quality research	LLM-rubric — tone/authenticity scoring from debate and interview coverage
Scandal typology & survival base rates	Scandal-typology and political-survival research	LLM-rubric classifies type/severity; deterministic lookup then matches it against real historical base rates for that scandal type from the archive
Down-ballot / ticket-splitting propensity	Split-ticket voting research	Deterministic — historical ticket-splitting rate per geography from <code>political_history</code>
Turnout / ground-game quality	Mobilization and GOTV research	LLM-rubric (field-office/GOTV mentions in coverage) + deterministic (early-vote request trend)
Domestic policy success/failure	Retrospective/performance voting, applied to legislative outcomes	LLM-rubric — salience/sentiment tied to Congress.gov bill-status ingestion
Foreign policy impact	Rally-round-the-flag and war-fatigue research	LLM-rubric — crisis salience and approval-direction scoring, a real but typically short-lived effect
Financial/market impact	Markets as a political-signal channel	Deterministic — the existing market-reaction ingestion, formalized as its own factor
Tech/media-environment impact	Algorithmic-amplification and virality research	LLM-rubric — narrative-virality and disinformation-risk scoring, used only to flag, never to suppress
Redistricting/competitiveness	Gerrymander-competitiveness research	Deterministic — the efficiency-gap/mean-median fairness scores (<i>Ahead of everything else</i>) feed straight in
Weather-and-turnout effects	Documented election-day weather/turnout correlation research	Deterministic — historical NOAA weather data correlated against turnout deltas
Third-party / spoiler dynamics	Spoiler-effect research in close races	Deterministic — third-party polling share against historical spoiler base rates at similar margins
Open-seat vs. incumbent-defended	Open-seat competitiveness research	Deterministic categorical flag with its own historically-calibrated base-rate adjustment
Endorsement cascades	Endorsement-effect research	Deterministic, weighted by a curated endorser-prominence score; the backtest determines which endorsement <i>types</i> actually correlate with movement

Every one of these lives in code as `modeling/factors_taxonomy.py` (the fixed rubric definitions) and `qualitative_factor_scores` (the scored, cited, cached data rows) — see [Schema & configuration](#).

10 — INTELLIGENCE

The Analyst

Ollama-powered, fed automatically by every demographic and historical fact the platform holds — reasoning, not inventing.

This is the "predictive genius" layer, and it's built the way GlobeGrid's grounded Q&A analyst was built: the model never gets to make a number up, because it never has to — it's handed a fully-assembled, cited context bundle before it writes a word. Ask it "why is this district tightening" or "what does the demographic trend in this county actually predict" and it reasons over real data, not vibes.

Ollama, local, by default. The same model already running on your machine drives race narratives, dossier synthesis, and every Analyst answer, with cloud providers wired in only as an optional fallback if Ollama isn't reachable. Nothing about the "genius" experience requires an API key or a monthly bill.

§ How the context gets assembled — automatically

Every sync job that lands new Census, FEC, Congress.gov, or certified-results data doesn't just update a table — it **invalidates** the standing **context pack** for every geography and race it touches, but does *not* eagerly rebuild it. A context pack is a deterministically-built bundle: the full demographic profile at every tier from precinct up through state, the historical result and turnout trend where it exists, the current poll average, fundamentals snapshot, and Qualitative Factor Scorecard ([The genius layer](#)), the fundraising trend, and any tagged rhetoric/news facts — everything the Analyst is allowed to reason over, nothing it has to guess. No LLM call happens until this bundle exists; if a tier's data is thin, the bundle says so honestly instead of the model quietly filling the gap.

§ Sized for a real machine, not a hypothetical one

Default model: Qwen3-30B-A3B-Instruct — a mixture-of-experts model with 30.5B total parameters but only 3.3B active per token, meaning it runs at roughly 8B-class speed on ordinary hardware despite its 262K-token context window, which is exactly the "large context, still fast locally" combination this feature needs. **Constrained-hardware fallback: Qwen3 14B** at 128K context, for machines without much RAM to spare. **Optional heavy upgrade: Llama 3.3 70B**, for anyone running a Mac Studio or a dual-GPU rig who wants the strongest reasoning available locally. All three are real, current Ollama-pullable models — this isn't aspirational.

A context pack targets roughly **28,000–32,000 tokens** — nowhere near any of the three models' actual context ceilings, deliberately, because stuffing a window to its limit adds latency and measurably dilutes attention quality even on long-context models. Budgeted roughly as: demographics across the relevant tiers plus comparison baselines (~4,000 tokens), historical trend (~4,000), current polling/fundamentals/forecast snapshot (~3,000), the Qualitative Factor Scorecard (~2,000), tagged news/rhetoric facts (~10,000 — the largest single allocation, and the one most likely to overflow), leaving headroom for conversation history and generation. When tagged facts alone would exceed their budget, the named policy is **relevance-ranked truncation**: facts are pre-scored by the same embedding-similarity mechanism the correlation engine already runs, the highest-relevance facts within budget are kept, and recency is the tiebreak only among facts of similar relevance — never a blind first-N or last-N cutoff.

Concurrency: Ollama serves one generation at a time per loaded model, so the real design problem is queueing, not parallelism. A two-lane priority queue solves it: *interactive* requests

(a live user mid-conversation with the Analyst) always preempt *background* requests (a context-pack rebuild), never the other way around. The rebuild-storm scenario a nightly sync could otherwise cause — tens of thousands of precincts all invalidated at once — never materializes in the first place, because rebuilds are **lazy, not eager**: invalidating a context pack only marks it stale; it's actually rebuilt the next time someone (a user or another feature) asks about that specific entity, and cached from then on. A precinct nobody has looked at simply stays stale and unbuilt indefinitely, at zero cost.

§ What it can actually do

Grounded Q&A

Free-text questions about any race, candidate, or geography, answered from that entity's context pack with inline citations back to the specific poll, filing, or demographic figure used.

Demographic deep-dive

Full profile of any precinct, county, district, or state, plus how its demographic trajectory compares to its state and national baseline.

Historical-analog reasoning

"Which past races looked like this one" — pulled from the correlation engine's historical-analog match against the archive back to 1950, narrated with the actual outcome each time.

Predictive read, clearly labeled

A plain-language synthesis of what the deterministic forecast, the fundamentals, the coalition detector, and the Qualitative Factor Scorecard are jointly pointing at — always followed by the real probability number, never replacing it.

WHY THIS ISN'T THE SAME THING AS LETTING THE MODEL FORECAST

The Analyst can hold more demographic and historical detail in a single answer than a human analyst could hold in their head — that's the genius part. What it can't do is move a single probability, average, or call by itself: every number in its context pack was computed before it was ever handed to the model, and every number it repeats back has to match what's already stored. It reasons brilliantly about facts it's forbidden to invent.

The LLM boundary

Stated once here, then enforced file by file everywhere it's used.

ALLOWED

Race-narrative prose (grounded, cited, never a call) · candidate/party dossier synthesis (background, cached, always over a curated floor) · rhetoric/topic-stance classification inside a fixed rubric · media-framing summaries · briefings · the Analyst's grounded Q&A and predictive reasoning · counterfactual branch narrative (each branch still scored by a deterministic precedent count) · **Qualitative Factor Scorecard scoring** — always against the fixed taxonomy, always cited, and the score itself is just a data row until the deterministic ensemble touches it.

NEVER ALLOWED, UNDER ANY CIRCUMSTANCE

Poll averaging · the fundamentals model · ideology scoring · forecasting math · the elastic-net weights that combine the Factor Scorecard into a probability · the Brier backtest that gates forecasts · correlation thresholds · the volatility index · anomaly detection · the race-calling decision itself.

Provider chain: local Ollama first (free, private, no rate limit), cloud providers as automatic fallback behind it. Background jobs skip a rate-limit cooldown instantly rather than competing with a live user request for quota. Every generative feature ships a fully-specified, non-blocking deterministic fallback — a null narrative with confidence forced low, an honestly-labeled non-AI counterfactual tree, a structured bulleted briefing — and the pipeline actually exercises that fallback path, not just handles it in theory.

12 — EXPERIENCE

The map

The same hand-rolled WebGL2 technique, the same tier system, aimed at counties instead of countries.

Tier 1 is a genuine 3D sphere rendered in raw WebGL2 — no three.js — defaulting to a locked, US-centered orthographic view with the rest of Earth rendered dim and non-interactive purely for visual context. Day/night terminator, city lights over population centers, cinematic fly-to camera between states and districts, particle-trail "correlation threads" linking related race-dynamics events (a fundraising spike and a poll swing in the same race, say) — the identical visual language that makes a conflict map feel alive, aimed at a district instead of a warzone.

§ Zoom LOD

National

50 states + DC, generalized boundaries, statewide poll averages and forecast shading.

State

Counties (and parish/borough equivalents) resolve, congressional districts toggle on as an independent overlay layer since district lines cut across county lines.

County-equivalent

Precinct/VTd detail where a state actually publishes it — best-effort, honestly flagged `thin_coverage` where it doesn't, exactly like a "thin coverage" badge on a sparse data region.

Historical

Congressional district lines are redistricting-versioned with `effective_from / effective_to`; the time-capsule scrubber can show the 2012-cycle map and the 2022-cycle map as genuinely different geometries, not the same lines relabeled.

§ Thematic modes & Election Night Mode

Choropleth modes: partisan lean (a computed PVI-style metric, never LLM-guessed), and — at whichever tier is on screen — any single demographic layer on its own: education attainment, race/ethnicity composition, median household income, age structure, urbanicity, foreign-born share. Plus polling average by race, forecast probability, the volatility index, turnout/early-vote intensity, fundraising density, ad-spend density. **Election Night Mode** is the War Mode trick pointed at a race instead of a conflict: click a contested race, the camera flies to the state or district's bounding box, counties fill in live as AP/state results certify, the feed splits into results/turnout/fundraising/media tabs, and the UI carries a visibly distinct state for `CALLABLE — awaiting human review` versus `CALLED by [named human]` — the human-only rule enforced visually, not just in the backend.

§ Offline build pipeline

Census TIGER/Line shapefiles (states, counties, congressional districts, VTDs where published) pulled once, simplified with Douglas-Peucker and quantize-then-polyline-encoded exactly like a boundary pipeline built for a whole planet — vendored as static frontend data

plus a self-contained backend point-in-polygon twin, so reverse-geocoding a poll or a news item to its county needs zero GIS dependency at runtime, ever.

13 — EXPERIENCE

Build-your-own map

The 270ToWin move, done on the same renderer that runs the live map.

Pick a race type and a cycle — Presidential, Senate, Governor, or House — and every unit on the map becomes clickable through a full outcome gradient: Safe / Likely / Lean for each party, Tossup, or Other, the same confidence language a real forecast uses. Click a state, a Senate seat, or a House district and it cycles; the topline number — 270 electoral votes, 51 Senate seats, 218 House seats — recomputes live, in plain arithmetic, off whatever's been assigned so far.

§ Scoped to how each office actually works

Presidential

State-by-state, except Maine and Nebraska, which the tool splits by congressional district plus the two statewide electors — the same `elector_method` field the live model reads, so a scenario is never wrong about how those two states actually work.

Senate

Only the seats actually up that cycle are toggleable; the rest display fixed at their current holder, exactly like a real Senate map has to.

House

All 435 districts are toggleable, defaulting to the competitive ones expanded and the safe ones collapsed to their projected holder, so the tool is usable without scrolling past 400 non-competitive seats first.

Governor

Every state with a race that cycle, nothing else.

§ Redistricting-aware, by construction

Build a scenario for a past cycle and the tool pulls *that cycle's* actual district lines from the same boundary-versioning system the live map uses — a 2012 what-if map is drawn on 2012 lines, not today's, because the two are genuinely different shapes.

§ Saving & export — entirely client-side

No account, no server round-trip, no new database table: a scenario is a compact per-unit outcome list that fits in a URL — and the math behind "fits" is worked out, not assumed. Each unit's outcome is one of eight states (Safe/Likely/Lean for each party, Tossup, or Other) — exactly 3 bits. The stress case is House: 435 districts $\times$ 3 bits = 1,305 bits $\approx$ 164 bytes; add a 4-byte header (schema version, race type, cycle year) and base64url-encode the result — `[168+3] $\times$ 4 = 224 characters`. That's the whole payload, comfortably inside any browser or server's practical URL-length ceiling with room to spare; President, Senate, and Governor scenarios are all under 60 units and encode to well under 30 characters. Saving locally works the same way GlobeGrid's camera bookmarks do — named, dated, kept in the browser, switchable between several at once. Sharing is a permalink carrying that exact encoded string. Exporting reuses GlobeGrid's snapshot trick exactly: crop the live renderer's own canvas, composite a title/legend/totals panel directly onto it, and download a PNG with zero server involvement — plus a plain CSV/JSON export of the raw assignments for anyone who wants to analyze it elsewhere.

DIFF AGAINST THE MODEL

An optional overlay compares a saved scenario to the platform's own live deterministic forecast and highlights exactly which units were called differently — a fast way to see where your assumptions and the model's actually disagree, unit by unit.

14 — EXPERIENCE

Polls, everywhere

One global window to browse every poll ever ingested — and the same data waiting on whatever entity you're already looking at.

A dedicated Polls window lists every individual poll the platform has ever ingested, filterable by race type, state or district, pollster, date range, sampled population (likely voters / registered voters / adults), and disclosed methodology. Every row carries the pollster's field dates, sample size, margin of error, topline numbers, the house-effect-adjusted number actually used in the average, and a direct link back to the pollster's own release — never a paraphrase, consistent with the primary-source rule everywhere else in this build.

The same list is embedded, pre-filtered, wherever it's relevant: open a state and see its Senate, Governor, and Presidential-in-that-state polls right there; open a House district and see that race's polls; open the national view and see the generic ballot. Nothing is hidden behind a separate tab you have to know to look for.

A POLL BELONGS TO THE RACE IT WAS FIELDIED FOR, NOT TO A CLICK

Pollsters field surveys at the national, state, or district level — essentially never the county level, and the platform doesn't pretend otherwise. Open a county page and you'll see the state's and district's polls that cover it, clearly labeled at that scope, with one click up to the full list for the broader race — never a fabricated "county poll" that was never actually taken.

Every pollster carries a transparent accuracy grade (detailed in [Ahead of everything else](#)) right next to its name in every list — so a reader can see at a glance whether a headline-grabbing outlier poll comes from a shop with a strong track record or a thin one.

15 — EXPERIENCE

Articles, everywhere

The news feeder doesn't wait for coverage to arrive — it goes and finds it, race by race, candidate by candidate.

A generic RSS firehose ([Ingestion layer](#)) will always miss the long tail — down-ballot House coverage, local-outlet reporting on a competitive statewide race, anything no national wire happens to pick up. Nobody curates a feed for "every article about NC-13." So PollGrid doesn't rely on one: it actively hunts.

§ The race search profile

Every tracked race carries a search profile, built the moment the race is seeded and kept current automatically: race-name variants (the official name, common shorthand, "{state} {office} race," "{state} {office} election {cycle}"), election-cycle tags, and every candidate's name variants — full legal name, common nickname, title-plus-surname ("Sen. Ossoff"), and alias forms pulled straight from FEC filing name variants. The moment a new candidate's official filing record lands — FEC or state SoS — their names are added to the profile automatically. A candidate gets searched for from the instant they're a real filer, not once they're already famous enough for a generic feed to stumble onto.

§ Retrieval — deliberately not GDELT

GDELT is excluded on purpose. GlobeGrid's own history is the reason why — it was retired after reportedly "trashing the system" with low-quality volume, and that lesson carries straight over. The default, free, keyless mechanism is per-query Google News RSS (news.google.com/rss/search?q={query}) run on the same interval discipline as any other source; a real commercial News API is wired in as an optional paid upgrade, mirroring the exact same free-first/paid-upgrade shape already used for Ollama-vs-cloud and native-feeds-vs-AP.

§ Dedup, relevance, and not becoming spam

Targeted-search hits land in the same `raw_items` table under the same `UNIQUE(source_id, external_id)` dedup as the generic firehose, so an article both paths find isn't double-counted — it just carries a `matched_profile_id` marker recording which race/candidate profile(s) matched, for traceability (one article can legitimately match several, e.g. two candidates in the same debate story). Nothing is filtered pre-ingestion on a relevance guess; instead, every hit flows through the identical extraction/correlation pipeline as everything else, and a genuinely irrelevant hit simply fails to correlate strongly and fades out of relevance-ranked views on its own. A profile whose query returns nothing for a stretch is logged and health-tracked exactly like a quiet source elsewhere in this design — never retried in a tight loop.

§ On every race and candidate page

An Articles panel — the same "everywhere" pattern as polls — sits on every race, candidate, and party page, pre-filtered to that entity, sorted by recency or relevance, each row showing the outlet, its reliability tier, and a direct link, never a paraphrase. Route:

```
/api/articles/{entity_type}/{id} .
```

Electoral history & the archive

Click a county, and it doesn't just show today's number — it shows every election it's ever been part of.

Every geographic entity — a state, a congressional district, a county-equivalent, a precinct where coverage allows — carries an Electoral History panel: every applicable race type, every past cycle, margin, turnout, and swing from the prior cycle, plotted as a simple deterministic trend. A state shows President by cycle plus both Senate classes plus Governor. A district shows House by cycle. A county shows how it voted in every race that touches it.

§ Boundary-aware, not boundary-blind

Congressional districts get redrawn. A district's history panel makes that visible instead of hiding it — a result from before a redistricting cycle is shown as belonging to *that* version of the district's geometry, with a clear marker at every redraw, never silently spliced onto today's lines as if the same label meant the same shape a decade earlier.

§ How deep the import actually goes

Stated per office, honestly, rather than one flat cutoff year — because the truth is different for each one:

OFFICE	IMPORT DEPTH	THE HONEST CAVEAT
President	Electoral votes & state popular vote from 1789	1789--1820 is flagged distinctly: many state legislatures chose electors directly rather than holding a popular vote. A clean statewide popular-vote series is reliable from 1824 forward.
Senate	Direct-election results from 1913/1914 (the 17th Amendment)	Appointments and special elections are their own tagged event type, never presented as an ordinary cycle.
House	District-level results as far back as a boundary-matched crosswalk exists	Strong and standardized from the mid-1970s forward; earlier cycles are imported where digitized sources exist and confidence-tagged accordingly, never smoothed to look uniformly complete.
Governor	Each state's own record	Same confidence-tagging discipline as the rest — a thin or uncertain row says so.

Every historical row carries a `confidence` tag applied once, at import time, and never upgraded retroactively just because a cleaner number would look better — the same provenance discipline GlobeGrid built for its own 1945-forward archive, run here across a genuinely longer and messier historical record.

Frontend & UI

Buildless, themeable, and genuinely sound-designed — the "coolness" part.

No React, no Vite, no npm, no CDN — native ES modules, a plain state object, a fixed bag of callback closures standing in for dependency injection, UI components as plain classes instantiated once and mutated directly. The same three-tier renderer (WebGL2 globe → 2D canvas → list) with a one-shot capability check, a manual override, and a graceful step-down if a tier's constructor ever throws. A single left-docked sliding pane hosts every detail view — race, candidate, party, geography at any tier — with real navigation history. A command palette (Ctrl/Cmd+K) jumps to any race, candidate, or geography by name. The live feed pushes correlated story clusters over a hand-rolled WebSocket, never raw articles, with a REST-polling fallback if the socket drops. The time-capsule scrubber replays the live feed and map for any moment since ingestion began; browsing a race's full electoral history further back than that is the archive's job, described above, not the scrubber's.

§ Themes

A handful of genuinely distinct palettes, each deriving the map's entire color treatment from one accent variable — no per-theme renderer color table to maintain. Deliberately **no default red-vs-blue partisan skin**: a neutral graphite-and-brass default (the identity of this very document), a "Broadcast Night" theme evoking election-night TV graphics, a light "Newsroom" mode, and a high-contrast/colorblind-safe mode.

§ Sound

Real Web Audio synthesis, not sample playback — an in-browser convolution reverb, composable presets, proper ADSR envelopes — ambient tone and tempo tracking the volatility index the way an earlier design tracked a conflict-instability score, plus distinct chimes for a poll movement, a breaking-news item, and a race call landing.

Ahead of everything else

Not by copying what RealClearPolitics, 270ToWin, Silver Bulletin, and Ballotpedia each do well — by making every one of those a byproduct of the same fact chain instead of five disconnected tools.

Qualitative signal, systematized

The [genius layer](#) extracts cited, rubric-scored qualitative factors — [two dozen and counting](#) — from the full information environment at a scale no analyst team could sustain, then lets a backtested statistical model, never the LLM itself, decide whether that signal earns its way into the number.

Coverage that hunts, not waits

[Targeted search profiles](#) per race and candidate actively find down-ballot and local coverage a generic firehose would simply never surface.

Chamber-control simulation

A Monte Carlo run across every individual race's own deterministic probability distribution, respecting cross-race correlation — a national-environment shock moves similar seats together, not independently — yields an actual "probability the Senate/House flips," not just a pile of separate seat odds.

A transparent pollster scorecard

Every pollster graded on real historical error against certified results, published openly, and that same grade — not a hidden internal number — is what sets its house-effect weight in the average.

A full audit trail on every number

Click any average or forecast and see the itemized computation: which polls, which weights, when it last moved and exactly why. Radical transparency instead of a black box you're asked to trust.

Redistricting fairness metrics, for free

Efficiency gap and mean-median difference, computed deterministically per district plan — nearly costless given the boundary history the map already carries, and something almost nobody in this space surfaces well.

Comprehensive candidate coverage

A real page for every candidate who's ever appeared on a general-election ballot for one of the four covered offices, seeded automatically the moment an official filing record creates the row — not just whoever's currently in the news.

Open data, not a walled garden

CSV/JSON/API export of any table, for anyone who wants to check the platform's own work or build something new on top of it.

THE ACTUAL MOAT

The map, the polls, the electoral history, the forecast, and the Analyst all read off one fact chain — so "why is this number what it is" always has exactly one real answer, traceable end to end, instead of three different tools quietly disagreeing and nobody able to say why.

Schema & configuration

Real column types, real constraints, a real config.yaml tree, and a testable definition of "Phase A is done."

§ config.yaml, concretely

Every key below is schema-validated at boot — wrong type, out-of-range value, or unknown key all fail the process loudly before `migrate()` ever runs:


```

ingestion:
  intervals_seconds:
    polls: 900          # 15 min
    fec: 3600
    census: 86400       # daily check, actual data changes far less often
    congress_gov: 3600
    news_rss: 120
    targeted_search: 600
    markets: 300
    social: 900
    results_native: 30  # tightens automatically on election night, see §12
  ap_elections:
    enabled: false      # flip true only once an AP account exists
    api_key_env: AP_ELECTIONS_API_KEY
  resilience:
    backoff_multiplier: 2.0
    max_backoff_seconds: 1800
    down_after_failures: 3

polling:
  averaging:
    recency_half_life_days: 14
    house_effect_adjustment: true
    min_sample_size: 300
    weight_by_sample_size: true

fundamentals:
  weights:              # must sum to 1.0, validator checks it
    incumbency: 0.20
    generic_ballot: 0.25
    economic_index: 0.20
    partisan_lean: 0.25
    fundraising_ratio: 0.10

correlation:
  same_window_similarity_threshold: 0.78
  historical_similarity_threshold: 0.85
  same_window_max_gap_hours: 72

forecasting:
  enabled: false        # per-category override earned via backtest, see §07
  brier_ceiling: 0.20
  min_graded_predictions: 20
  auto_enable_earned: true

genius_layer:           # see §08
  ensemble_method: elastic_net_logistic
  refit_interval_days: 1
  context_pack_token_budget: 30000

llm_provider:
  primary: ollama_local
  fallback_order: [groq_free, anthropic, openrouter]
  ollama:
    default_model: qwen3:30b-a3b-instruct
    fallback_model: qwen3:14b
    upgrade_model: llama3.3:70b
    host: http://localhost:11434

demographics:
  sync_cadence_days: 30
  min_population_threshold: 0

```

§ Real DDL for the geography tier — the primary-key question, settled

Recommendation: use the Census Bureau's own GEOID/FIPS codes as **primary keys** wherever one already canonically exists (states, counties, congressional districts) rather than inventing a surrogate integer — it makes every join against Census/TIGER data direct with no separate mapping layer to maintain. Precincts get a surrogate **INTEGER PRIMARY KEY** instead, because no federal agency defines a canonical precinct ID; a state's own precinct code, where one exists, is kept as a plain column.

```

CREATE TABLE states (
  fips_code      TEXT PRIMARY KEY,      -- 2-digit Census FIPS, e.g. '06'
  usps_code      TEXT NOT NULL UNIQUE,  -- 'CA'
  name           TEXT NOT NULL,
  elector_method TEXT NOT NULL CHECK (elector_method IN ('winner_take_all','congressional_district')),
  electoral_votes INTEGER NOT NULL,
  is_territory   INTEGER NOT NULL DEFAULT 0
);

CREATE TABLE congressional_districts (
  geoid          TEXT NOT NULL,        -- Census GEOID, e.g. '0601'
  congress_number INTEGER NOT NULL,    -- which Congress this boundary version was seated for
  state_fips      TEXT NOT NULL REFERENCES states(fips_code),
  district_number INTEGER NOT NULL,    -- 0 = at-large
  effective_from  TEXT NOT NULL,      -- ISO date
  effective_to    TEXT,               -- NULL = current
  PRIMARY KEY (geoid, congress_number)
);

CREATE TABLE county_equivalents (
  geoid          TEXT PRIMARY KEY,      -- 5-digit Census FIPS, e.g. '06037'
  state_fips      TEXT NOT NULL REFERENCES states(fips_code),
  name           TEXT NOT NULL,
  type           TEXT NOT NULL CHECK (type IN ('county','parish','borough','census_area','independent_city'))
);

CREATE TABLE precincts (
  precinct_id     INTEGER PRIMARY KEY,
  county_geoid    TEXT NOT NULL REFERENCES county_equivalents(geoid),
  state_precinct_code TEXT,
  name           TEXT,
  coverage_confidence TEXT NOT NULL CHECK (coverage_confidence IN ('measured','derived','thin_coverage')),
  source_tier     TEXT NOT NULL CHECK (source_tier IN ('native','vest','openelections','manual'))
);

CREATE TABLE demographics (
  id              INTEGER PRIMARY KEY,
  tier            TEXT NOT NULL CHECK (tier IN ('nation','state','congressional_district','county_equivalent','precinct')),
  entity_id      TEXT NOT NULL,        -- polymorphic: the relevant tier's own PK
  as_of          TEXT NOT NULL,
  category       TEXT NOT NULL CHECK (category IN ('population_age','race_ethnicity','education','economic','housing_urbanicity')),
  variable       TEXT NOT NULL,        -- e.g. 'median_household_income'
  value          REAL,
  confidence     TEXT NOT NULL CHECK (confidence IN ('measured','derived')),
  source         TEXT NOT NULL,        -- e.g. 'acs5_2024:B19013_001E'
  UNIQUE (tier, entity_id, as_of, category, variable)
);

CREATE TABLE political_history (
  id              INTEGER PRIMARY KEY,
  tier            TEXT NOT NULL CHECK (tier IN ('state','congressional_district','county_equivalent','precinct')),
  entity_id      TEXT NOT NULL,
  office         TEXT NOT NULL CHECK (office IN ('president','senate','governor','house')),
  cycle_year     INTEGER NOT NULL,
  winner_party   TEXT,
  margin_pct     REAL,
  turnout_pct    REAL,
  confidence     TEXT NOT NULL CHECK (confidence IN ('measured','derived','uncertain')),
  source         TEXT NOT NULL,
  UNIQUE (tier, entity_id, office, cycle_year)
);

```

"PHASE A IS DONE" — A TESTABLE DEFINITION, NOT A FEELING

All 56 rows present in `states` (50 + DC + 5 territories), each with a valid `elector_method`. Exactly 435 current-vintage rows in `congressional_districts` (`effective_to` IS NULL). `county_equivalents` row count between 3,140 and 3,150 (the real-world count fluctuates slightly with consolidations — treating 3,143 as a rigid target is itself a bug). Zero rows returned by `SELECT geoid FROM county_equivalents WHERE state_fips NOT IN (SELECT fips_code FROM states)` — the referential-integrity check that must pass before Phase A is considered complete. And `config.yaml` validates clean against the schema above with zero warnings.

§ The rest of the schema, grouped

GROUP	TABLES
Core fact chain	<code>sources</code> , <code>raw_items</code> , <code>polls</code> , <code>extracted_facts</code> (hash-chained, append-only)
Geography & demographics	<code>states</code> , <code>county_equivalents</code> , <code>congressional_districts</code> , <code>precincts</code> , <code>demographics</code> , <code>political_history</code> — real DDL above
Political entities	<code>candidates</code> (every historical filer, not just current), <code>officeholders</code> , <code>parties</code> , <code>pacs</code> , <code>donors_aggregated</code> , <code>media_outlets</code>
Races	<code>racess</code> , <code>primaries</code> , <code>race_candidates</code> , <code>electoral_vote_allocations</code> , <code>redistricting_events</code>
Modeling	<code>poll_averages</code> , <code>fundamentals_snapshots</code> , <code>ideology_scores</code> , <code>forecasts</code> , <code>chamber_simulations</code> , <code>volatility_scores</code> , <code>anomaly_flags</code> , <code>second_order_links</code> , <code>coalition_models</code>
The genius layer	<code>qualitative_factor_scores</code> (entity + factor_key + score + citation + <code>as_of</code> , one row per re-score), <code>ensemble_weights</code> (the elastic-net coefficients, refit per category, stored as data), <code>ensemble_backtest_results</code> (qualitative-augmented vs. quantitative-only Brier comparison, per category)
Articles & retrieval	<code>race_search_profiles</code> (name variants, cycle tags, candidate aliases, auto-updated on new filings), <code>article_entity_links</code> (which race/candidate profile(s) an ingested article matched)
Rhetoric, framing & spend	<code>media_framing_scores</code> (outlet + story + topic + framing-direction, rubric-scored), <code>topic_stance_scores</code> (candidate + topic + stance, rubric-scored), <code>ad_spend</code> (race + sponsor + medium + amount, sourced from FEC independent-expenditure filings)
Trust & transparency	<code>pollster_ratings</code> (historical error, grade, feeds the averaging weight), <code>redistricting_fairness_scores</code> (efficiency gap, mean-median difference, per district plan), <code>computation_audit_log</code> (the itemized trace behind every average/forecast)
The Analyst	<code>context_packs</code> (per entity, invalidated on upstream sync, rebuilt lazily on demand — see §10), <code>analyst_sessions</code> , <code>analyst_messages</code>
Predictions & provenance	<code>predictions</code> (hash-chained), <code>backtest_results</code> , <code>correlation_feedback</code>
Election night	<code>results_live</code> (+ <code>source_tier</code> : native/openelections/ap/manual), <code>race_calls</code> (<code>called_by</code> NOT NULL, human only)
Engagement	<code>watchlist_items</code> , <code>bookmarks</code> , <code>annotations</code> , <code>daily_briefings</code>

Deliberately not in this list: no `map_scenarios` table — a build-your-own-map scenario is client-side state encoded straight into its share URL, never server-stored. And `ballot_measures` / any state-legislative `chamber_composition` table are intentionally absent too — races stay typed generically enough to add both later, but nothing seeds, syncs, or serves either until state houses are back in scope.

API surface

One route module per concern, every read route taking `as_of` .

ROUTE FAMILY	COVERS
<code>/api/races</code> , <code>/api/races/{id}</code>	Race detail: averages, fundamentals, forecast, narrative, corroboration
<code>/api/candidates</code> , <code>/api/parties</code>	Curated-floor + synced + AI-filled dossiers
<code>/api/demographics/{tier}/{entity_id}</code>	Full demographic panel for one state/district/county-equivalent/precinct, with confidence tags
<code>/api/demographics/trends/{race_id}</code>	The coalition detector's output — which demographic variables explain the most movement
<code>/api/polls</code>	Global filterable poll browser — race type, geography, pollster, date range, methodology
<code>/api/articles/{entity_type}/{id}</code>	Every article a race/candidate/party's search profile has matched, recency- or relevance-sorted
<code>/api/factors/{race_id}</code>	The Qualitative Factor Scorecard for a race — every factor, its score, and its citation
<code>/api/forecast/ensemble/{race_id}</code>	Quantitative-only vs. qualitative-augmented probability, side by side, plus which one is actually live for that category
<code>/api/races/{id}/framing</code>	The media-framing matrix and ad-spend-by-sponsor breakdown for a race
<code>/api/entities/{tier}/{id}/history</code>	Electoral history for one state/district/county-equivalent/precinct, across every applicable race type, boundary-aware
<code>/api/pollsters/{id}/rating</code> , <code>/api/pollsters/ratings</code>	The transparent accuracy scorecard, and what it sets the house-effect weight to
<code>/api/districts/{id}/fairness</code>	Efficiency gap and mean-median difference for a district plan
<code>/api/forecast/chamber/{chamber}</code>	Monte Carlo chamber-control probability (Senate/House majority)
<code>/api/audit/{metric_id}</code>	The itemized computation trace behind one average or forecast number
<code>/api/map/*</code>	Boundary tiles by LOD, thematic mode values (incl. per-demographic-category choropleths), live pins
<code>/api/electionnight/*</code>	Live results, <code>CALLABLE</code> flags, human call submission
<code>/api/counterfactual</code>	Scenario generation and on-demand branch deepening
<code>/api/analyst/query</code>	Grounded Q&A over an entity's context pack, Ollama-first
<code>/api/forecast/scorecard</code>	The Brier backtest dashboard — every category, including failing ones
<code>/api/export/{table}</code>	Open CSV/JSON export of any table — no walled garden
<code>/ws/feed</code>	Live story-cluster push, hand-rolled WebSocket, REST-poll fallback
<code>/api/config</code> , <code>/api/diagnostics</code>	Mirrored tunables; a page that actually executes every AI feature live

Directory layout

PATH	WHAT'S THERE
config.yaml	Every tunable — nothing hardcoded elsewhere
core/	Config loader, DB session/locking, provenance hash chain, gazetteer
ingestion/	Scheduler + one adapter per source family, plus results/tier1_native/ , results/tier2_openelections.py , results/tier3_ap.py , results/manual_entry.py , and targeted_search.py (the race-search-profile hunter, §15)
modeling/	Averaging, fundamentals, ideology, forecasting, chamber_simulation, correlation, volatility, coalition_detector, counterfactual, corroboration, rhetoric, race_calling, pollster_ratings, redistricting_fairness, audit_trail, genius_ensemble.py (the elastic-net stacking model, §08), factors_taxonomy.py (the fixed rubric definitions, §09)
domain/	Candidates (every historical filer), parties, races, geography (states → districts → county-equivalents → precincts) and demographics at every tier, the curated-floor knowledge layer
analyst/	Context-pack assembly (lazy rebuild, priority-queued), the Ollama-first reasoning/Q&A engine, cloud-fallback provider chain
api/	Router, route modules, WebSocket feed
frontend/	Buildless SPA — map tiers, the map builder (fully client-side), a global polls window, an articles panel on every entity page, race/candidate/party/geography pages with electoral-history panels, live feed, the Analyst, time capsule
scripts/	Offline boundary/redistricting build pipeline, seeds, per-office historical backfill importers (President to 1789, Senate to 1913, House/Governor as deep as records allow)
tests/	python -m unittest discover

Build sequence

Lettered phases, dependency-ordered — the same convention both prior projects used, because it works.

A Repo scaffold, config loader + schema validator against the real <code>config.yaml</code> tree, five-tier geography seed against the real DDL — both in §19 — with the stated referential-integrity check as the actual done-criterion	B Demographic ingestion pipeline — the concrete ACS endpoints and table IDs from §05, precinct-level derivation via named population-weighted areal interpolation, honest confidence tagging	C Parties, candidates, officeholders schema + curated-floor seed for the current cycle
D Races, primaries, race-candidate linkage, electoral-vote allocation model, historical race archive back to 1950	E Poll ingestion adapters (first dozen-plus pollsters) + normalization	F Poll averaging engine (deterministic)
G FEC finance ingestion	H Fundamentals model, now demographic-aware	I Ideology scoring (deterministic proxy)
J Forecasting engine + the Brier backtest gate	K Prediction-market ingestion + forecast/market divergence	L Campaign-event & transcript ingestion + rhetoric/topic taxonomy
M Correlation engine (same-window + historical-analog against the archive back to 1950 landed in phase D — deepened later in phase AA)	N Demographic coalition detector, volatility index, anomaly detection	O Live feed + WebSocket
P Map Tier 1 (WebGL2) + Tiers 2/3 + offline boundary build pipeline, five tiers, redistricting-aware	Q Thematic map modes, including per-demographic-category choropleths at every tier	R The Analyst — context-pack assembly + Ollama-first reasoning & Q&A
S Race/candidate/party/geography pages, SlidePane shell, command palette	T Counterfactual engine	U Corroboration engine
V Tier-1 native state-results ingestion + <code>CALLABLE</code> flagging + human-call UI — tiers 2/3 (§05) land in phase NN	W Provenance hash chain + verification CLI	X Themes, sound design, accessibility pass
Y Time-capsule scrubber over the live feed and map	Z Ops hardening — backups, API budgets, diagnostics page, nightly jobs	

A-through-Z is the platform. The second wave below is what makes it the best version of itself — comprehensive history, comprehensive candidates, and the features nobody else in this space does well.

AA Full historical archive import — President to 1789, Senate to 1913, House/Governor as deep as digitized crosswalks allow, every row confidence-tagged by era	BB Comprehensive candidate backfill — a real page for every historical candidate across all four offices, seeded from archival filing records	CC Electoral-history panel on every entity page, boundary-aware across redistricting cycles
DD Global polls window + contextual embedding on every race/geography page	EE Pollster accuracy scorecard — historical grading, feeding both the public badge and the averaging house-effect weight	FF Build-your-own map — the scenario tool, client-side, URL-encoded sharing, canvas PNG export
GG Chamber-control Monte Carlo simulation	HH Redistricting fairness metrics (efficiency gap, mean-median difference) per district	II Public audit trail — the itemized computation behind every average and

	plan	forecast number
JJ Open data export — CSV/JSON/API access to any table A third wave — the qualitative-genius layer and its supporting cast — deliberately lands last, because every phase in it depends on infrastructure the first two waves already built: the deterministic forecast to compare against, the Analyst's LLM-calling plumbing to reuse, and a real historical archive deep enough to backtest a new model against. KK Targeted news retrieval — race search profiles, per-query ingestion (§15), the Articles-everywhere panel LL The factors taxonomy (§09) implemented — every deterministic formula wired to real columns, every LLM-rubric factor scoped and prompted MM The genius ensemble (§08) — Qualitative Factor Scorecard scoring pipeline, the elastic-net stacking model, and its mandatory backtest-vs-quantitative-baseline gate before anything qualitative-augmented is ever shown live NN Official-results tiers 2/3 — bulk OpenElections/VEST sync for historical depth, the optional gated AP adapter for anyone ready to pay for it		
Draft architecture, written for review before anything lands in the repo — nothing here has been committed to <code>CLAUDE.md</code> , scaffolded, or scheduled. Say the word and phase A starts.		